

Harry Nguyen

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EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • GPA: 3.75/4.0

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award
- Activities: Amazon University Event – BEN Tech Cloud Connect ATL

EXPERIENCE

TiMoto AI | Software Engineer

Sep 2025 – Present

- Led backend, cloud infrastructure, and ML serving for a 3-person engineering team — designed, deployed, and operated **distributed production systems** end-to-end
- Designed and maintained **gRPC inter-service layer with exactly-once semantics**; resolved production deadlock under concurrent calls by tracing shared resource acquisition conflicts and redesigning call sequencing — **100% evaluation success rate** at **sub-50ms p99**
- Architected **vLLM inference engine with PagedAttention** for LLM serving — eliminated KV cache memory fragmentation under concurrent load; deployed with continuous batching achieving **zero OOM failures** at production traffic
- Architected multi-AZ ECS Fargate with **Terraform IaC**, CloudWatch observability, and circuit breaker pattern — **99.9% uptime**, **44% cost reduction** (\$40–60/month); auto-rollback on health check failure
- Maintained distributed production systems across gRPC, PostgreSQL, and ML serving layers — triaged and resolved incidents via root-cause analysis, documented runbooks, and applied post-mortem learnings; participated in on-call rotation

Google | Software Engineer Intern — Chrome Browser

May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with Protocol Buffers — selected Protocol Buffers over custom serialization for schema evolution and cross-language compatibility; shipped to Chrome stable channel serving **3B+ active users** at **sub-50ms p99**, **10K+ req/sec**
- Identified settings navigation as p99 bottleneck at 1,200ms; designed and implemented **lock-free concurrent trie search** eliminating mutex contention — **96% latency reduction**, zero production regressions
- Architected event-driven TypeScript/React system with observer pattern decoupling UI state propagation — accelerated feature delivery **68%** across **25K+ lines** of Chromium at **95% test coverage**
- Collaborated with Chrome infrastructure team to deliver design documents and code reviews; changes adopted into production branch reviewed by senior Chrome engineers

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless BaaS on Amazon Web Services (AWS) — selected JWT over session auth for horizontal scalability; auto-scaling policy supporting **500+ concurrent users**, **90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+ on large datasets; redesigned with PostgreSQL indexing achieving **sub-100ms** for 10,000+ records

PROJECTS

Pulumi — Contributor | Go, TypeScript, Infrastructure-as-Code (IaC)

github.com/pulumi

- Submitted Go CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning — under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer — studied linearizability guarantees and correctness under concurrent operations and partial failures

SKILLS

Distributed Systems: gRPC, Protocol Buffers, Exactly-once semantics, Circuit breakers, Fault tolerance, Multi-AZ, Kafka, Event-driven architecture

Languages: C++, Go, Python, TypeScript, Java, JavaScript, SQL, Bash, Rust

Cloud & Infrastructure: AWS (ECS Fargate, EKS, EC2, S3, RDS), Terraform, Kubernetes, Docker, CI/CD, CloudWatch, Prometheus, Grafana, GCP

ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LoRA/QLoRA, PyTorch, LangChain, LLM-as-a-judge evaluation

Frameworks & Databases: Django, FastAPI, Node.js, React, REST APIs, PostgreSQL, Redis, MongoDB

AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor